

Multiple Choice (10 marks)

1. Statement 1| The original ResNets and Transformers are feedforward neural networks. Statement 2| The original Transformers use self-attention, but the original ResNet does not.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
2. The model obtained by applying linear regression on the identified subset of features may differ from the model obtained at the end of the process of identifying the subset during
 - (a) Best-subset selection
 - (b) Forward stepwise selection
 - (c) Forward stage wise selection
 - (d) All of the above
3. Statement 1| Layer Normalization is used in the original ResNet paper, not Batch Normalization. Statement 2| DCGANs use self-attention to stabilize training.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
4. Existential risks posed by AI are most commonly associated with which of the following professors?
 - (a) Nando de Freitas
 - (b) Yann LeCun
 - (c) Stuart Russell
 - (d) Jitendra Malik
5. Which of the following statements about Naive Bayes is incorrect?
 - (a) Attributes are equally important.
 - (b) Attributes are statistically dependent of one another given the class value.
 - (c) Attributes are statistically independent of one another given the class value.
 - (d) Attributes can be nominal or numeric

True / False (10 marks)

Answer True or False

6. True or False: The computational complexity of gradient descent is linear in the number of iterations performed.
7. True or False: As the number of training examples approaches infinity, the variance of the model's predictions decreases.
8. True or False: Dropout increases the activation values of neurons to prevent overfitting in neural networks.
9. True or False: The use of weak classifiers helps prevent overfitting when performing bagging in machine learning.
10. True or False: To calculate $P(H|E, F)$, the sufficient numbers include $P(E, F)$, $P(H)$, and $P(E, F|H)$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to find the sum of even numbers at even positions.
12. Write a function to find the index of the last occurrence of a given number in a sorted array.

End of Paper